

PRODUCT DOCUMENTATION

Nutrition Facts Label Generator

User guide & technical documentation — build a recipe from ingredients, set the serving, and get a professional US FDA-style Nutrition Facts label with the official Daily Values and FDA rounding rules applied automatically.

Creator	**Dr. Asif Ahmad**
Product	Nutrition Facts Label Generator — version 1.0
Category	Health
Published by	Asif SaaS Studio
Date	20 October 2026
Format	Single-file web application (index.html, 32.4 KB)

Private by design: recipes never leave the user's device — no uploads, no accounts, no tracking. And honest by design: the ingredient values are reference values for drafting and education; a commercial label must rest on laboratory analysis or a verified database (Chapter 9).

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1. What this product does — and who it serves

Nutrition Facts Label Generator turns a recipe into the classic black-and-white US Nutrition Facts label. You pick ingredients and amounts; the app totals the nutrients, divides them per serving, applies the FDA Daily Values (21 CFR 101.9) and the FDA label rounding rules, and renders the label — ready to export as a crisp PNG, an SVG with real text, or a printable HTML page. A detail table beside the label shows the exact, unrounded figures per serving and per 100 g, so you can always see what the rounding did.

Audience	Typical use
Small food producers & home businesses	Drafting a compliant-style label for a new product before paying for laboratory analysis.
Students & teachers	Food science and nutrition classes: recipe formulation, Daily Values, rounding rules, label reading.
Recipe developers & bloggers	Publishing a per-serving nutrition panel alongside a recipe.
Anyone curious	Seeing what a homemade mix — a trail mix, a smoothie, a sweetened drink — actually contains per serving.

2. The workflow, step by step

Step	What you do	What happens
1 · Serving information	Type the serving size exactly as it should print (e.g. 1 pack (50 g)) and the servings per container; tick about if the count is approximate.	These become the label's servings lines, verbatim.
2 · Build the recipe	Add ingredients from the database dropdown and set each amount in grams. Edit or remove rows with ×.	The recipe total weight and nutrient totals update live.
3 · Add custom ingredients	Enter a name and any nutrient values from your own data sheet (milk powder, protein powder, your flour blend…).	The ingredient joins the recipe and stays in the dropdown for reuse in the session.
4 · Read the label	Watch the label and the exact-value detail table.	Per-serving values are FDA-rounded on the label; the table shows exact per-serving and per-100 g figures.
5 · Export	Download PNG (3×), SVG, or printable HTML.	Files are generated in the page — nothing is uploaded anywhere.

A one-click **sample recipe** (trail mix) is loaded on first open, so the label is never blank and every control can be tried immediately.

3. The ingredient database

The built-in database holds **38 reference ingredients**. Every value is stated **per 100 g** of ingredient and covers 15 nutrients: energy (kcal), total fat, saturated fat, trans fat, cholesterol, sodium, total carbohydrate, dietary fibre, total sugars, added sugars, protein, vitamin D, calcium, iron and potassium. The values are **USDA-style standard reference values** — the familiar published figures for each food — chosen for drafting, teaching and product development (see Chapter 9 for what that does and does not authorise).

Group	Ingredients (38)
Dairy & eggs	Whole milk · Skim milk · Cheddar cheese · Butter · Yogurt, plain · Greek yogurt, nonfat · Egg, whole
Oils	Olive oil · Canola oil
Meat, poultry & fish	Chicken breast, cooked · Ground beef 90%, cooked · Salmon, cooked
Grains, bread & pasta	White rice, cooked · Brown rice, cooked · White bread · Whole wheat bread · Rolled oats · All-purpose flour · Spaghetti, cooked · Sweet corn
Fruit	Banana · Apple · Dates
Vegetables	Potato · Sweet potato · Tomato · Onion · Spinach · Broccoli · Carrots
Legumes & soy	Lentils, cooked · Black beans, cooked · Tofu, firm
Nuts & spreads	Almonds · Peanut butter
Sweeteners & pantry	Granulated sugar · Honey · Cocoa powder, unsweetened

3.1 Custom ingredients

Any ingredient not in the list can be added from its data sheet — a supplier specification, a packet label, or your own analysis. Custom ingredients follow the same **per-100 g convention**: every value you type is the amount of that nutrient in 100 g of the ingredient, exactly as printed on most data sheets. Leave fields blank for nutrients you do not know; they count as zero (the validation rules in Chapter 10 require a name and at least one value, and all values must be numbers of 0 or more).

3.2 Why per 100 g?

Per-100 g is the one convention every food database shares, which is what makes the maths in Chapter 4 a single multiplication: an ingredient used at g grams contributes value × g ÷ 100 of each nutrient. It also matches how the EU and most other systems publish reference data, so values can be copied from almost any source without conversion.

4. The maths — from recipe to label

Three steps, all exact until the final rounding:

1. **Recipe totals.** For each nutrient, the app sums per-100 g value × grams used ÷ 100 across all ingredients, and totals the recipe weight. Rows with an amount of 0 g or less are ignored by the maths (and flagged by validation).
2. **Per serving.** Each recipe total is divided by the servings-per-container figure. This exact value drives the label.

3. **Per 100 g.** Each recipe total is also expressed per 100 g of finished product (total ÷ recipe weight × 100) for the detail table — the figure other countries' formats are built from.

Only after these steps does rounding happen, and rounding is applied per nutrient, on the per-serving value, using the FDA rules in Chapter 5. % Daily Value is computed from the unrounded per-serving value against the Daily Value, then rounded — to the nearest whole percent for the macronutrients, and through the FDA declaration tiers for vitamins and minerals.

5. The FDA framework — Daily Values and rounding

The label follows the current US rules (2016 revision, codified at 21 CFR 101.9). Two tables do all the work.

5.1 Daily Values (for a 2,000 kcal daily diet)

Nutrient	Daily Value	Nutrient	Daily Value
Total fat	78 g	Dietary fibre	28 g
Saturated fat	20 g	Added sugars	50 g
Cholesterol	300 mg	Protein	50 g
Sodium	2300 mg	Vitamin D	20 mcg
Total carbohydrate	275 g	Calcium	1300 mg
		Iron	18 mg
		Potassium	4700 mg

Trans fat and total sugars have no Daily Value, so their %DV column is blank on the label — exactly as the FDA format requires. Protein has a Daily Value (used in the detail table), but its %DV is not printed on the label; see Chapter 7.

5.2 Rounding rules implemented

Quantity	Rule
Calories	Under 5 → **0** · up to 50 → nearest **5** · above 50 → nearest **10**
Total / saturated / trans fat	Under 0.5 g → **0** · up to 5 g → nearest **0.5 g** · above 5 g → nearest **1 g**
Carbohydrate, fibre, sugars, protein	Under 0.5 g → **0** · under 1 g → **1 g** · otherwise nearest **1 g**
Cholesterol	Under 2 mg → **0** · otherwise nearest **5 mg**
Sodium	Up to 5 mg → **0** · up to 140 mg → nearest **5 mg** · above 140 mg → nearest **10 mg**
Vitamin & mineral %DV	Declared in tiers: under 2% → **0%** (or 2% if any is present) · up to 10% → nearest **2%** · up to 50% → nearest **5%** · above 50% → nearest **10%**

6. Two worked examples, verified

Both examples were computed end-to-end by the shipped code and checked by hand; they are also part of the automated test suite (Chapter 12).

6.1 Sweetened milk — 50 g whole milk + 25 g granulated sugar, 1 serving

Label line	Shows	Label line	Shows
Calories	**130**	Total Sugars	27 g
Total Fat	1.5 g · 2%	Includes Added Sugars	25 g · 50%
Saturated Fat	1 g · 5%	Protein	2 g
Cholesterol	5 mg · 2%	Vitamin D	0.5 mcg · 4%
Sodium	20 mg · 1%	Calcium	57 mg · 4%
Total Carbohydrate	27 g · 10%	Iron	0 mg · 0%
Dietary Fiber	0 g · 0%	Potassium	70 mg · 2%

Reading the rounding at work: the exact energy is 127.3 kcal per serving, which the calories rule (>50 → nearest 10) prints as **130**; the exact fat is 1.65 g, printed as 1.5 g by the 0.5 g step; the 25 g of sugar is almost pure added sugar, hence 50% of the 50 g Daily Value on its own.

6.2 Trail mix (the in-app sample) — 80 g almonds + 60 g dates + 60 g rolled oats, 4 servings of 1 pack (50 g)

Label line	Shows	Label line	Shows
Calories	**220**	Total Sugars	11 g
Total Fat	11 g · 14%	Includes Added Sugars	0 g · 0%
Saturated Fat	1 g · 5%	Protein	7 g
Cholesterol	0 mg · 0%	Vitamin D	0 mcg · 0%
Sodium	0 mg · 0%	Calcium	72 mg · 6%
Total Carbohydrate	25 g · 9%	Iron	1.6 mg · 10%
Dietary Fiber	5 g · 18%	Potassium	320 mg · 8%

The 200 g batch carries 862.8 kcal in total — **431.4 kcal per 100 g** in the detail table — and the dates contribute all of the sugars yet zero added sugars, because their sugar is intrinsic to a whole fruit (Chapter 7). Halving the servings to 2 doubles the label to 430 calories, a useful classroom demonstration that serving size is a decision, not a measurement.

7. Anatomy of the label

The generator renders the standard FDA **vertical format**: the heavy "Nutrition Facts" title; servings-per-container and serving-size lines under a thick rule; the large Calories line; the % Daily Value column; the nutrient rows in the FDA's fixed order with their hierarchy of rules (thick, medium, hairline); and the footnote explaining % Daily Value against a 2,000-calorie diet. Three details are worth knowing:

- **Protein %DV is omitted.** US labels declare protein in grams but print a %DV only when a protein claim is made or the food is for infants/children. The generator follows that standard practice — protein's %DV still appears in the detail table for reference.
- **"Includes Added Sugars"** counts sugars added during processing — granulated sugar, honey, syrups, and the sugar fraction of sweetened ingredients — following the FDA definition. Sugars naturally present in whole foods (the lactose in milk, the fructose in dates or banana) are not added sugars; they appear only in Total Sugars. In the database, each ingredient's added-sugars field encodes this distinction, and custom ingredients ask for it explicitly.
- **Indentation is meaning.** Saturated and trans fat sit under Total Fat; fibre and sugars sit under Total Carbohydrate; Added Sugars sits one level deeper under Total Sugars — the visual hierarchy of the official format is preserved in all three renderers (page, canvas, SVG).

8. Exports

Format	How it is produced	Best for
PNG (3×)	The label model is drawn on a canvas at three times display size (1020 px wide), so text stays crisp in documents and slides.	Documents, presentations, product mock-ups.
SVG	A hand-built vector file whose text is real text (not outlines), laid out from the same label model at 340 px width.	Print workflows, design tools, scaling without loss.
Printable HTML	A clean standalone page containing only the label, generated in the browser; print it or save it as PDF from the browser's print dialog.	Paper copies, classroom handouts, PDF via print.

All three exports come from one shared label model, so the page preview, PNG, SVG and printout can never disagree with each other — a property the test suite checks directly.

9. Accuracy, legality and honest scope

> **Reference values are for drafting and education.** A label printed on a product you sell must be based on **laboratory analysis of your product** or a **verified database entry for your exact ingredients and process**. Ingredient composition varies with variety, season, supplier and cooking — reference values cannot capture that, and regulators (and customers) are entitled to the real figures.

- **Format vs. data.** The generator applies the FDA format, Daily Values (21 CFR 101.9) and rounding faithfully. What it cannot certify is your data — that part is yours, and Chapter 6's examples show exactly how far reference data goes.
- **US format only.** The label is the United States format. Other jurisdictions differ — the EU uses a per-100 g table, Pakistan follows PSQCA rules — and the per-100 g column in the detail table is provided to help with those, not to replace them.
- **No hidden AI or lookup.** Version 1.0 contains no database lookup and needs no API key. A future verified-data version could query **USDA FoodData Central** — but that service requires an API key, and keys must never sit inside a web page. The honest architecture is a small **backend proxy** holding the key, which the page calls over HTTPS; that is a possible v2, and it is not faked in v1.
- **Allergens, claims and exemptions** (nutrient-content claims, allergen declarations, small-business exemptions) are legal matters outside this tool's scope — take advice before commercial use.

10. Validation rules

Field	Rule	What you see
Ingredient amount	Must be greater than 0 g.	A red warning — "Every ingredient needs an amount greater than 0 g." Invalid rows are excluded from the maths, never silently treated as correct.
Servings per container	Must be greater than 0.	A red warning and no division by zero — per-serving values fall back safely.
Custom ingredient — name	Required.	"Give your ingredient a name first."
Custom ingredient — values	Every value a number of 0 or more; at least one value entered.	"Nutrient values must be numbers of 0 or more." / "Enter at least one nutrient value (per 100 g)."
Database integrity	38 ingredients × 15 nutrient values, none negative.	Checked by the automated test suite (Chapter 12), not left to chance.

11. Publishing to Wix CMS

Field	Value
Title	Nutrition Facts Label Generator
Category	Health
Slug	nutrition-label-generator
Summary	Turn recipes into FDA-style Nutrition Facts labels — 38-ingredient reference database, custom ingredients, official Daily Values & FDA rounding, PNG/SVG/ printable exports.
Version	1.0 · Demo: No

1. Wix dashboard → CMS → **SaasProducts** → Add Item.
2. Fill the listing fields above; paste the full contents of `index.html` into **appHtml**.
3. Set **published** on, save, then open the live app page and run the testing checklist in the test report.

12. How correctness was verified

The shipped core (Appendix) was exercised by **50 automated checks — all passed**. Highlights:

- **Every rounding rule**, at its boundaries: calories 4 → 0, 37 → 35, 137 → 140; fat 0.4 → 0, 3.24 → 3 g, 7.6 → 8 g; gram nutrients 0.7 → 1; cholesterol 13 → 15 mg; sodium 147 → 150 mg; vitamin/mineral tiers 23% → 25% and 67% → 70%.
- **%DV maths**: 39 g fat = 50% of 78 g; 1150 mg sodium = 50% of 2300 mg.
- **Two complete recipes end-to-end** (Chapter 6): sweetened milk prints Calories 130 with Added Sugars 25 g · 50%; the trail-mix sample prints Calories 220 with Total Fat 11 g · 14%, and 431.4 kcal per 100 g in the detail figures.
- **Edge behaviour**: rows with 0 g or negative grams are skipped safely; the label model is complete at 21 rows in FDA order; the SVG generator emits the full label; the database itself is validated — 38 ingredients, 15 values each, none negative.

13. Product information and credits

Field	Detail
Product	Nutrition Facts Label Generator
Version	1.0 (20 October 2026)
Creator	**Dr. Asif Ahmad**
Publisher	Asif SaaS Studio
Category / slug	Health / nutrition-label-generator
Deliverable	index.html — 32.4 KB, single file, all CSS and JavaScript inline
Requirements	Any modern browser; no server, no installation, no account; no localStorage, service workers, permissions or parent-window access
Verification	50/50 automated checks against the shipped core, including two hand-verified recipes (Chapter 12)

Appendix — the tested core, verbatim

The code below is the complete working core of the product exactly as it ships inside `index.html`: the 38-ingredient database, the FDA Daily Values, the rounding rules, the recipe maths, the label model and the SVG renderer. Interface wiring is omitted.

```
/* Nutrient field order used everywhere:
   kcal, fat, satFat, trans(g), chol(mg), sodium(mg), carbs, fiber, sugars, addedSugars, protein,
   vitD(mcg), calcium(mg), iron(mg), potassium(mg) — all per 100 g. */
```

```
var NUTRIENT_KEYS =
["kcal","fat","satFat","trans","chol","sodium","carbs","fiber","sugars","added","protein","vitD","calcium","iron
","potassium"];
var INGREDIENTS = [
  ["Whole milk",            [61,3.3,1.9,0.1,10,43,4.8,0,5.0,0,3.2,1.0,113,0.03,132]],
  ["Skim milk",             [34,0.1,0.06,0,2,42,5.0,0,5.1,0,3.4,1.0,122,0.03,156]],
  ["Cheddar cheese",        [403,33,19,0.9,105,621,1.3,0,0.5,0,25,0.6,721,0.14,76]],
  ["Butter",                [717,81,51,3.3,215,11,0.1,0,0.1,0,0.9,1.5,24,0.02,24]],
  ["Olive oil",             [884,100,13.8,0,0,2,0,0,0,0,0,0,1,0.56,1]],
  ["Canola oil",            [884,100,7.4,0,0,0,0,0,0,0,0,0,0,0]],
  ["Egg, whole",            [143,9.5,3.1,0,372,142,1.1,0,0.4,0,12.6,2.2,56,1.75,138]],
  ["Chicken breast, cooked",[165,3.6,1.0,0,85,74,0,0,0,0,31,0.1,15,1.04,256]],
  ["Ground beef 90%, cooked",[217,10,4.0,0.4,88,72,0,0,0,0,26,0.1,18,2.6,318]],
  ["Salmon, cooked",        [208,13,3.1,0,55,59,0,0,0,0,20,12.4,9,0.34,363]],
  ["White rice, cooked",    [130,0.3,0.08,0,0,1,28,0.4,0.05,0,2.7,0,10,0.2,35]],
  ["Brown rice, cooked",    [112,0.9,0.18,0,0,5,24,1.8,0.4,0,2.6,0,10,0.53,86]],
  ["White bread",           [265,3.2,0.7,0,0,491,49,2.7,5.0,4.0,9.0,0,115,3.6,115]],
  ["Whole wheat bread",     [247,3.4,0.7,0,0,399,41,7.0,4.4,4.0,12.9,0,107,2.5,254]],
  ["Rolled oats",           [389,6.9,1.2,0,0,2,66,10.6,0.3,0,16.9,0,54,4.72,429]],
  ["All-purpose flour",     [364,1.0,0.16,0,0,2,76,2.7,0.3,0,10.3,0,15,1.2,95]],
  ["Granulated sugar",      [387,0,0,0,0,1,100,0,99.9,99.9,0,0,1,0.05,2]],
  ["Honey",                 [304,0,0,0,0,4,82,0.2,82,82,0.3,0,6,0.42,52]],
  ["Banana",                [89,0.3,0.1,0,0,1,23,2.6,12.2,0,1.1,0,5,0.26,358]],
  ["Apple",                 [52,0.2,0.03,0,0,1,14,2.4,10.4,0,0.3,0,6,0.12,107]],
  ["Potato",                [77,0.1,0.03,0,0,6,17,2.2,0.8,0,2.0,0,12,0.81,421]],
  ["Sweet potato",          [86,0.1,0.02,0,0,55,20,3.0,4.2,0,1.6,0,30,0.61,337]],
  ["Tomato",                [18,0.2,0.03,0,0,5,3.9,1.2,2.6,0,0.9,0,10,0.27,237]],
  ["Onion",                 [40,0.1,0.04,0,0,4,9.3,1.7,4.2,0,1.1,0,23,0.21,146]],
  ["Spinach",               [23,0.4,0.06,0,0,79,3.6,2.2,0.4,0,2.9,0,99,2.71,558]],
  ["Broccoli",              [34,0.4,0.04,0,0,33,6.6,2.6,1.7,0,2.8,0,47,0.73,316]],
  ["Carrots",               [41,0.2,0.04,0,0,69,9.6,2.8,4.7,0,0.9,0,33,0.3,320]],
  ["Almonds",               [579,50,3.8,0,0,1,21.6,12.5,4.4,0,21.2,0,269,3.71,733]],
  ["Peanut butter",         [588,50,10,0,0,459,20,6.0,9.2,8.0,25.1,0,43,1.9,649]],
  ["Yogurt, plain",         [61,3.3,2.1,0,13,46,4.7,0,4.7,0,3.5,0.1,121,0.05,155]],
  ["Greek yogurt, nonfat",  [59,0.4,0.1,0,5,36,3.6,0,3.2,0,10,0,110,0.07,141]],
  ["Lentils, cooked",       [116,0.4,0.05,0,0,2,20,7.9,1.8,0,9.0,0,19,3.33,369]],
  ["Black beans, cooked",   [132,0.5,0.14,0,0,1,23.7,8.7,0.3,0,8.9,0,27,2.1,355]],
  ["Tofu, firm",            [76,4.8,0.7,0,0,7,1.9,0.3,0.3,0,8.1,0,350,2.7,146]],
  ["Spaghetti, cooked",     [158,0.9,0.18,0,0,1,31,1.8,0.6,0,5.8,0,7,1.3,44]],
  ["Sweet corn",            [96,1.5,0.2,0,0,15,21,2.4,4.5,0,3.4,0,2,0.52,270]],
  ["Dates",                 [277,0.2,0,0,0,1,75,6.7,66.5,0,1.8,0,64,0.9,696]],
  ["Cocoa powder, unsweetened",[228,13.7,8.1,0,0,21,57.9,37,1.8,0,19.6,0,128,13.86,1524]]
];
/* FDA Daily Values (2016 label rules, 21 CFR 101.9) */
var DV = { fat:78, satFat:20, chol:300, sodium:2300, carbs:275, fiber:28, added:50, protein:50, vitD:20,
calcium:1300, iron:18, potassium:4700 };
function dvPct(key, value){
  if (!DV[key]) return null;
  var p = value / DV[key] * 100;
  return Math.round(p);
}
/* ------- FDA rounding conventions (simplified to the label rules) ------- */
function roundTo(x, step){ return Math.round(x / step) * step; }
function roundCalories(v){
  if (v < 5) return 0;
  if (v <= 50) return roundTo(v, 5);
  return roundTo(v, 10);
}
function roundFat(v){ /* total, saturated, trans */
  if (v < 0.5) return 0;
  if (v <= 5) return roundTo(v, 0.5);
  return roundTo(v, 1);
}
function roundGram(v){ /* carbs, fiber, sugars, protein */
  if (v < 0.5) return 0;
  if (v < 1) return 1;
  return roundTo(v, 1);
}
function roundChol(v){ if (v < 2) return 0; return roundTo(v, 5); }
function roundSodium(v){
  if (v <= 5) return 0;
  if (v <= 140) return roundTo(v, 5);
  return roundTo(v, 10);
}
function roundMicroPct(p){ /* vitamin/mineral %DV declaration tiers */
  if (p < 2) return p <= 0 ? 0 : 2;
  if (p <= 10) return roundTo(p, 2);
```

```
  if (p <= 50) return roundTo(p, 5);
  return roundTo(p, 10);
}
function fmtG(v){ /* label grams: whole or .5 */
  return (Math.round(v * 2) / 2).toString().replace(/\.0$/, "");
}
/* ------- Recipe maths -------
   items: [{values:[15 per-100g numbers], grams}] */
function computeRecipe(items){
  var totals = {}, i, k;
  NUTRIENT_KEYS.forEach(function(key){ totals[key] = 0; });
  var weight = 0;
  items.forEach(function(it){
    var g = Number(it.grams);
    if (!(g > 0)) return;
    weight += g;
    NUTRIENT_KEYS.forEach(function(key, idx){ totals[key] += it.values[idx] * g / 100; });
  });
  return { totals: totals, weight: weight };
}
function perServing(totals, servings){
  var out = {};
  NUTRIENT_KEYS.forEach(function(key){ out[key] = servings > 0 ? totals[key] / servings : 0; });
  return out;
}
function per100g(totals, weight){
  var out = {};
  NUTRIENT_KEYS.forEach(function(key){ out[key] = weight > 0 ? totals[key] / weight * 100 : 0; });
  return out;
}
/* ------- Label model (shared by HTML / canvas / SVG renderers) -------
   Row: {l, lv, r, bold, indent, size, rule} rule: 0 none, 1 thin, 2 medium, 3 thick (rule AFTER the row)
   Special kinds: title, servings, servingSize, calHead, calValue, dvHead, foot */
function buildLabelModel(ps, meta){
  var rows = [];
  function pct(key, v, micro){
    var p = dvPct(key, v);
    if (p === null) return "";
    if (micro) p = roundMicroPct(p);
    return p + "%";
  }
  rows.push({ kind:"title", text:"Nutrition Facts" });
  rows.push({ kind:"servings", text:(meta.about ? "about " : "") + meta.servings + " servings per container" });
  rows.push({ kind:"servingSize", l:"Serving size", r:meta.servingSize, rule:3 });
  rows.push({ kind:"calHead", text:"Amount per serving", rule:0 });
  rows.push({ kind:"calValue", l:"Calories", r:String(roundCalories(ps.kcal)), rule:2 });
  rows.push({ kind:"dvHead", text:"% Daily Value*", rule:1 });
  rows.push({ l:"Total Fat", lv:fmtG(roundFat(ps.fat)) + "g", r:pct("fat", ps.fat), bold:true, rule:1 });
  rows.push({ l:"Saturated Fat", lv:fmtG(roundFat(ps.satFat)) + "g", r:pct("satFat", ps.satFat), indent:1, rule:
1 });
  rows.push({ l:"Trans Fat", lv:fmtG(roundFat(ps.trans)) + "g", r:"", indent:1, italic:true, rule:1 });
  rows.push({ l:"Cholesterol", lv:roundChol(ps.chol) + "mg", r:pct("chol", ps.chol), bold:true, rule:1 });
  rows.push({ l:"Sodium", lv:roundSodium(ps.sodium) + "mg", r:pct("sodium", ps.sodium), bold:true, rule:1 });
  rows.push({ l:"Total Carbohydrate", lv:fmtG(roundGram(ps.carbs)) + "g", r:pct("carbs", ps.carbs), bold:true,
rule:1 });
  rows.push({ l:"Dietary Fiber", lv:fmtG(roundGram(ps.fiber)) + "g", r:pct("fiber", ps.fiber), indent:1, rule:
1 });
  rows.push({ l:"Total Sugars", lv:fmtG(roundGram(ps.sugars)) + "g", r:"", indent:1, rule:0 });
  rows.push({ l:"Includes Added Sugars", lv:fmtG(roundGram(ps.added)) + "g", r:pct("added", ps.added), indent:2,
rule:1 });
  rows.push({ l:"Protein", lv:fmtG(roundGram(ps.protein)) + "g", r:"", bold:true, rule:2 });
  rows.push({ l:"Vitamin D", lv:(Math.round(ps.vitD * 10) / 10) + "mcg", r:pct("vitD", ps.vitD, true), rule:
1 });
  rows.push({ l:"Calcium", lv:roundTo(ps.calcium, 1) + "mg", r:pct("calcium", ps.calcium, true), rule:1 });
  rows.push({ l:"Iron", lv:(Math.round(ps.iron * 10) / 10) + "mg", r:pct("iron", ps.iron, true), rule:1 });
  rows.push({ l:"Potassium", lv:roundTo(ps.potassium, 10) + "mg", r:pct("potassium", ps.potassium, true), rule:
0 });
  rows.push({ kind:"foot", text:"* The % Daily Value (DV) tells you how much a nutrient in a serving of food
contributes to a daily diet. 2,000 calories a day is used for general nutrition advice." });
  return rows;
}
/* ------- SVG label from the model ------- */
var LABEL_W = 340;
function rowHeight(row){
  if (row.kind === "title") return 44;
  if (row.kind === "calValue") return 40;
  if (row.kind === "foot") return 52;
```

```
  if (row.kind === "servings" || row.kind === "servingSize") return 21;
  if (row.kind === "calHead") return 17;
  if (row.kind === "dvHead") return 17;
  return 20;
}
function ruleWidth(rule){ return rule === 3 ? 11 : rule === 2 ? 5 : rule === 1 ? 1 : 0; }
function labelHeight(rows){
  var h = 20;
  rows.forEach(function(r){ h += rowHeight(r) + ruleWidth(r.rule || 0); });
  return h;
}
function svgEsc(s){ return String(s).replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;"); }
function buildLabelSVG(rows){
  var H = labelHeight(rows), parts = [];
  parts.push('<svg xmlns="http://www.w3.org/2000/svg" width="' + LABEL_W + '" height="' + H + '" viewBox="0 0 '
+ LABEL_W + ' ' + H + '">');
  parts.push('<rect x="1" y="1" width="' + (LABEL_W - 2) + '" height="' + (H - 2) + '" fill="#FFFFFF"
stroke="#000000" stroke-width="2.5"/>');
  var y = 12;
  function text(x, yy, str, size, weight, anchor, style){
    parts.push('<text x="' + x + '" y="' + yy + '" font-family="Arial, Helvetica, sans-serif" font-size="' +
size + '" font-weight="' + weight + '" text-anchor="' + anchor + '"' + (style ? ' font-style="' + style + '"' :
"") + '>' + svgEsc(str) + '</text>');
  }
  rows.forEach(function(r){
    var rh = rowHeight(r);
    if (r.kind === "title"){ text(10, y + 32, r.text, 31, 900, "start"); }
    else if (r.kind === "servings"){ text(10, y + 15, r.text, 13.5, 400, "start"); }
    else if (r.kind === "servingSize"){ text(10, y + 15, r.l, 14.5, 700, "start"); text(LABEL_W - 10, y + 15,
r.r, 14.5, 700, "end"); }
    else if (r.kind === "calHead"){ text(10, y + 13, r.text, 12.5, 700, "start"); }
    else if (r.kind === "calValue"){ text(10, y + 27, r.l, 21, 900, "start"); text(LABEL_W - 10, y + 33, r.r,
33, 900, "end"); }
    else if (r.kind === "dvHead"){ text(LABEL_W - 10, y + 13, r.text, 12, 700, "end"); }
    else if (r.kind === "foot"){
      var words = r.text.split(" "), lines = [], line = "";
      words.forEach(function(w){ if ((line + " " + w).length > 62){ lines.push(line); line = w; } else line =
line ? line + " " + w : w; });
      if (line) lines.push(line);
      lines.forEach(function(ln, i){ text(10, y + 13 + i * 12.5, ln, 10.5, 400, "start"); });
    } else {
      var x = 10 + (r.indent === 1 ? 16 : r.indent === 2 ? 32 : 0);
      var label = r.l + (r.lv ? "  " + r.lv : "");
      text(x, y + 15, label, 13.5, r.bold ? 700 : 400, "start", r.italic ? "italic" : null);
      if (r.r) text(LABEL_W - 10, y + 15, r.r, 13.5, 700, "end");
    }
    y += rh;
    var rw = ruleWidth(r.rule || 0);
    if (rw > 0){ parts.push('<rect x="10" y="' + y + '" width="' + (LABEL_W - 20) + '" height="' + rw + '"
fill="#000000"/>'); y += rw; }
  });
  parts.push("</svg>");
  return parts.join("\n");
}
```